recent guidebooks have paved the way for this by spreading knowledge and knowhow on useful methodologies and the necessary computational diagrams behind them.[99]

As already pointed out in this chapter, the dominating role of textual sources in the digital humanities faces growing criticism, much as it does in digital history. Before having a closer look at visual and audio-visual materials in the next chapter, it is worth recalling that the supposed duality between textuality and visuality is quite misleading and should not be taken too strictly. Texts include images and vice versa. Today, Internet texts often include both sound and visuals. And, moreover, both texts and images can be considered as material remains of the past. Text is not an abstract order of characters but is always, to draw on Bruno Latour's idea, inscribed on material objects, on stone or marble, on paper or hard drives.[100] Texts are material entities that have dimensions: printed texts manifest themselves, for example, as folios and octavos, as broadsheets and tabloids. When considering texts as big data, metatext must seriously be taken into account, since it offers access to the material dimensions of the texts at hand. Metadata has recently been successfully used in the study of book history, in which it is vital to comprehend how books circulated as concrete artefacts in the past.[101] In the future, these material ramifications might be avenues for bringing the quantifiable data of texts more closely together with the study of their contents and, thus, their impacts on the imaginations of readers.

3

Mapping and Viewing History

In the early stages of digital history, in the 1990s, maps were of particular interest, perhaps more so than the textual materials discussed in the previous chapter, because large-scale digitization projects had only recently begun. At the University of Virginia in 1993, William G. Thomas and Edward L. Ayers started a project on the US Civil War, entitled *The Valley of the Shadow*, that brought together the experiences of Confederate and Union soldiers. It published historical documents online, including letters, diaries, images, newspapers, church records, census records, tax records, statistics – and maps. The pre-war material, for example, included maps pertaining to geography, infrastructure, agriculture, politics, religion and slavery in the selected counties (Augusta in Virginia and Franklin in Pennsylvania). These were not merely digitized pre-existing visualizations but also an integral part of the project, produced for online use by the project members.[1]

Visual sources have been at the heart of digital history from the start, and maps have been an essential ingredient in visualizing archival data and establishing points for comparison. The following chapter continues by discussing maps further as sources and historical interpretations, fuelled by the so-called spatial turn. The chapter proceeds by selecting three other themes under discussion, the increased role of visual sources, including the idea of distant viewing,

recent developments in the multimodal analysis of audio-visual sources and the possibilities of metadata for visual and audio-visual history.

Maps as digital sources

Historians have always emphasized the meaning of space and spatiality. In the study of the past, *where* has been as important as *when*. This involves, for example, locations (such as cities and villages), natural landmarks (such as rivers, mountains or seas) and both natural and political borderlines. In addition to maps, spatiality has also been studied through an array of sources, from architectural plans and landscape images to verbal descriptions in travel accounts and fiction. Historians have also theorized on the significance of space, as demonstrated in Frederick Jackson Turner's famous essay 'The significance of the frontier in American history' (1893) and Fernand Braudel's classic *The Mediterranean and the Mediterranean World in the Age of Philip II* (1949).

The so-called spatial turn in the humanities and social sciences had its roots in this enduring interest in spaces and places,[2] but in the 1990s it really entered the scholarly discourse. According to Doris Bachmann-Medick, this development was inspired by the end of the Cold War at the turn of the 1980s and 1990s; with the end of the political polarity of Eastern and Western blocs, a need arose to map the reorganization of the global order, to go beyond previous security strategies and to understand how transnational flows would change the way the world was imagined.[3] Edward W. Soja's *Postmodern Geographies* was a groundbreaking volume in 1989, followed by his *Thirdspace* in 1996. Soon, the spatial turn embraced several fields of study, including anthropology, archaeology, geography, gender studies and history.[4] For the digital historian, it is important to be aware of these backgrounds, but of course spatiality and its digital research methods are strongly linked to the development of geospatial information that has emerged at the same time as a result of multidisciplinary collaboration.

The digitization of historical sources described in chapter 1 has also included maps. Today, many national and international collections of digitized maps enable their study using both computational tools and more traditional methods.[5] For example, the Royal Library of Denmark had already begun scanning its map collection in 1997. At first, the idea was to make index maps to demonstrate the holdings and increase the usability of the collection. Another motivation was the challenge presented by oversized maps that were difficult to handle, and, of course, preservation concerns were also paramount because the use of digital copies would allow the originals to remain untouched. Since 2003, the library has offered rare maps through the Internet.[6] The National Library of Argentina now has 1,122 downloadable historical maps on its website, and the digital collection features not only older maps but also a substantial amount of material from the twentieth century.[7] One of the largest current collections of historical maps, plans and views, comprising 4.5 million items, is located at the British Library. It includes not only scanned maps but also geospatial data, cartographic applications and digital aerial photographs, and it features both digitized and born-digital content.[8] The chronological range of the collection is over 2,000 years, and its size is suggested by the example of the King George III Topographical and Maritime collection, which alone includes '30–40,000 maps, plans and views, both printed and hand-drawn, of all parts of the world'.[9] Not all maps are two-dimensional, however, and the British Library also holds hundreds of terrestrial and celestial globes, which are three-dimensional material artefacts.[10]

As sources, maps are layered cultural artefacts that include numbers, texts, symbols and images. Historical research draws on these sources, which can be objects of careful close reading and iconographic analysis or can be used for more distant reading via the computational study of visual concepts or textual content. Maps can be described as representations of the physical world, but they also construct the world as we know it. Historical maps must always be understood in the context of the practices that originally contributed to their production, including the ways of measuring and representing distances and the strategies for aggregating data for maps; of course, the motives and uses of maps also influence the interpretation of these documents.

Maps possess a strong representational character as they usually try to capture the three-dimensional world in only two dimensions. In *Exploring Big Historical Data*, Shawn Graham, Ian Milligan and Scott Weingart point out that 'these seemingly straightforward representations are loaded with difficult choices, as laying two-dimensional coordinates onto a 3D world means making complicated choices around what map projection to use'.[11] They note, for example, that, in the Mercator projection that underlies Google Maps, Greenland looks almost as big as Africa, although Africa is, in reality, fourteen times larger.[12]

Present-day projections differ in how they address the problem of 2D/3D conversion, and many other factors influenced how old maps projected the physical world. In the past, maps also represented a realm of imagination, and their environmental features, such as coastlines and riverbeds, may have transformed over time, so it is not always possible to reduce historical maps to the topographical world. The researcher must estimate the possibility of matching a historical map with a modern one and determine whether this serves the purpose of the scholarly task. If so, the next step is trying to identify the points on the historical map that correspond to latitudes and longitudes in the present-day reference map (a process called *georectification*). Since the 1980s, geographical information systems (GIS) have provided tools for managing geographical information. GIS makes it possible to draw on historical maps by extracting their information and analysing them in comparison with other GIS data and data sets.[13] The New York Public Library's collection of over 11,000 georectified historical maps is openly accessible to users.[14]

Maps as historical interpretations

The use of historical maps (and in fact any source) depends on the research problem. Sometimes, georectification is not an option, for example in the case when no actual historical map is available. Other possible sources of topographical and geographical information can be further processed into a digital map, presenting a historical interpretation.

The historian Harri Kiiskinen drew on GIS in his study of the production and trade of *terra sigillata* pottery in Roman Etruria. To understand the trading routes, he tried to determine which rivers in Etruria were navigable at the time, 2,000 years ago. In addition to archaeological and historical evidence, this work used a digital terrain model based on topographic data from NASA's Shuttle Radar Topography Mission. As a result, Kiiskinen was able to draw a map that reconstructed the river network in Etruria.[15] Here, the digital map serves research purposes and expresses an important historical argument.

Maps had already been used to visualize big historical data before the era of GIS and the digital humanities. An excellent reminder of this is the production of Charles-Joseph Minard (1781–1870), a French civil engineer who won fame for his informational graphics and maps.[16] His most well-known map, from 1869, describes Napoleon's Russian campaign of 1812. There is a geographic orientation in the image of the direction of the campaign, but the map incorporates many other elements of information, from the number of troops to the temperature.[17] The map clearly shows how Napoleon's army suffered during its march towards Moscow from the Russian border. Minard also captured phenomena of his own era, for example, migratory movements on a global scale. The nineteenth century was similar to today in that it was characterized by a sudden abundance of data provided by new technologies, such as the high-speed press, the railroad and the telegram.[18] Today, Minard is often recognized as the developer of the so-called *flow map*; in 1845, spurred by the discussion of the development of railroad connections, he produced a map that showed how passengers circulated in the road routes between Dijon and Mulhouse.[19]

Nineteenth-century statistics have had a remarkable influence on how we visualize information. Graham, Milligan and Weingart mention the *dot density map* that was used, for example, by the English physician John Snow (1813–1858), who, drawing on statistical methods, revealed the correlation between contaminated water and cholera. Snow marked the locations of London cholera cases with a dot on a map and was able to show that many cases occurred around a water pump.[20]

Although recognizing these roots is important, digital technologies obviously offer many new avenues for maps, mapping and data visualization in general. Digital maps enable zooming in and out so that they can function as platforms for accumulated data for various historical sources. For example, a city map can include textual information and images of historical places for the user. The map can also be a channel for simultaneously presenting diverse research results. Minard did the same in his infographics, but digital maps are not as entangled with the two-dimensional world. Virtual maps can represent a third dimension and thus provide a broader visual perspective on the past.

From visual evidence to distant viewing

Thus far, this chapter has dealt with maps and mapping, which have been at the heart of digital history since the 1990s. What must be discussed next is actual visual evidence from the past, including how it can contribute to and change our understanding of history. Today, digital historians have access to manuscript illustrations, photos, posters and works of art, but digitization rates vary in collections that sometimes offer free use and sometimes only restricted use. For many reasons, the boundary between free and restricted use also varies. For example, there may be copyright concerns that influence the availability of images. In museum collections, copyright holders are not only artists but also photographers who have produced digital reproductions.

One of the most well-known examples of making visual content available is the Rijksmuseum in Amsterdam, which is a Dutch national museum that holds a collection of over a million objects, including 2,000 paintings. In 2012, the Rijksmuseum started an extraordinary project by making 125,000 high-resolution images available and downloadable. The plan is to add 40,000 images annually until the entire collection is online. Users can explore these images through the museum's own Rijksstudio, make observations, take clips and use them both in research and for other purposes. At the time of writing (May 2020), there were as many as 676,693 high-resolution images available through Rijksstudio.[21] The

collection is a substantial window, not only onto the Dutch past but also onto art history in general.

Restricted collections can also be useful for research because their metadata are often available online. An excellent example is provided by RMN Photo. RMN refers to the French organization Réunion des musées nationaux, a joint enterprise of French art galleries and museums. Today, RMN Photo also includes foreign galleries, such as the Prado in Madrid and the National Galleries of Scotland in Edinburgh.[22] The database includes small preview images that allow users to peek at the collection, and it is also possible to search by keyword. The collection is immense; a search of the keyword *arbre* (tree), returns over 22,000 images of various kinds of trees, including paintings, drawings, photographs, statues and many other kinds of objects from the Middle Ages to the present. This includes, for example, over 1,400 paintings from the nineteenth century.

In addition to RMN Photo and Amsterdam's Rijksmuseum, numerous national and international image collections can be put under the lens. National libraries around the world have digitized cultural artefacts and images, and many heritage projects offer visual content to users, from the American Memory project to Europeana. The latter, for example, contains transnational collections, such as First World War newsreels, postcards and letters. The film collection alone comprises as many as 2,726 items that can be regarded as one of the most comprehensive cinematic collections on the Great War.[23] It is obvious that these collections set different premises and conditions for research than textual material, but they also offer many possibilities for digital historians.

In his research lab Software Studies Initiative, Lev Manovich and his team have processed massive collections of images and video, including 6,000 paintings by French Impressionists and 20,000 photographs from the photography collection of the Museum of Modern Art. From a historical perspective, the Impressionist collection of images is comprehensive, as it represents approximately 50 per cent of all paintings and pastels presented in the Impressionist exhibitions during the years 1874–1886. The experiment of the Software Studies Initiative started with the idea of extracting particular features – in fact, 200 of them – from

the data set. This included 'colour characteristics, contrast, shapes, textures and some aspects of composition'.[24] These features were then computationally reduced to a smaller number of dimensions to enable the grouping of images and thus the viewing of them from a distance, as a whole. As a result, they found that the types of images that are usually associated with Impressionism, with lighter tones, were a smaller part of the data set. As Manovich writes, 'At least half of the images turn out to be rather traditional and more typical of classical 19th century painting (darker tones and warm colors).'[25]

Manovich presented this experiment in his article 'Data science and digital art history', published in 2015. The actual aim of the text was to articulate basic concepts for the study of large image corpora, drawing on data science and especially considering what the extraction features, such as average brightness and saturation, could offer to the digital analysis of large data sets of images. In 2019, Taylor Arnold and Lauren Tilton emphasized strongly what they call *distant viewing*:

> [W]e argue that DH [digital humanities] should consider, what we call, distant viewing – a methodological and theoretical framework for studying large collections of visual material. Distant viewing is distinguished from other approaches by making explicit the interpretive nature of extracting semantic metadata from images. In other words, one must 'view' visual materials before studying them. Viewing, which we define as an interpretive action taken by either a person or a model, is necessitated by the way in which information is transferred in visual materials.[26]

Manovich did not use the expression 'distant viewing', but this was exactly the aim of his texts: to try to track down what machine viewing could mean and how it could be used in the digital humanities. In recent years, 'distant viewing' has been employed by other researchers as well,[27] which shows the increasing interest in the computational study and cultural analytics of images. In the case of distant reading, the

question is: what does it mean when a machine 'reads' a text, which features have to be computationally extracted and how could these features be used for analysis? The same goes for the analysis of images, both still and moving: what does 'viewing' mean? This is what Arnold and Tilton underline in their article: there is a 'need for coding visual materials prior to exploratory data analysis of visual corpora'.[28] At the same time, it is important to recognize that there are different genres and different kinds of images; therefore, there are particular characteristics that the interpretation should take into account.[29]

Sound and vision

Parallel to the previous discussion, in data science, and not always in connection to the digital humanities, much research has taken place in relation to sound and vision. Automatic speech recognition (ASR) is today a useful tool for the study of audio-visual material. It enables the extraction of speech and its convergence into text, which again can be analysed with text mining tools. Simultaneously ASR processing can be deployed for producing digital indexes of the contents of auditive or audio-visual sources and thus help to enhance existing metadata.

Considering how much audio material is in the archives, computational methods could be employed to increase their accessibility and, of course, to study *en masse* sources that would otherwise be difficult to digest. As a research example, we could play with the idea of applying ASR to newsreels after the breakthrough of sound cinema in the late 1920s and early 1930s. Often, these newsreels were made with a silent camera, without any sound recorded on location. A voice-over narration was added later in the studio. Newsreels were a text-driven form of audio-visual culture; therefore, speech extraction might offer illuminating results. In any case, the quality of sound, and possible background music, would make this research challenging. Another problem might be the fact that ASR draws heavily on language models in identifying words. It seems that there is insufficient research on historical language models, which from a historian's

perspective would be essential for recognizing speech acts of the past. Not only language but also the way we speak, our enunciation, has changed tremendously. If we think of fictitious material, such as narrative films from the 1930s and 1940s, the dialogue might also include different dialects and slang. Furthermore, the quality of sound technology and the sounds of the filming location would make the recognition even more painstaking. Still, the idea to be able to mine and to distant-view old cinematic content is tempting, and it would offer an avenue into the imagination of past decades. There are sophisticated technologies that can further help to refine methodologies, such as audio event detection, voice activity detection and speaker diarization, which can help to identify who actually speaks and even connect the voice to the speaker when he or she is not visible on screen.[30]

Another area of recent exploration is the visual analysis of movie content, drawing on the technology around deep neural networks (DNNs). This includes, for example, such research strategies as visual feature and facial expression extraction and visual content description, which go beyond visuality into the multimodal analysis of audio-visual sources.[31] The development of content detectors helps to identify details in big audio-visual data sets. If the research project spans a long period, the analysis of movie content would give raw material, such as on how the environments, milieus and cultural artefacts used in these works change over time.

In the study of audio-visual content, it is necessary to pay attention to the changing technologies after the nineteenth century: they influence the ways in which these cultural products can be used for digital analysis and how they can be disseminated. Early film history was preserved on nitrate cellulose material which was very fragile and flammable. Therefore, many of the early films have been lost, and their study must be based on other kinds of sources, including stills, manuscripts, newspaper ads and reviews. Acetate cellulose is more stable and became dominant after the Second World War. Film archives around the world have for decades aimed at preserving these copies, first by producing acetate duplicates, now by making digital scans with 4k or even higher resolution. After the advent of video tape recorders in the 1950s, many other technologies for audio-visual products

have been introduced. In the 1980s Betamax and VHS technology dominated consumer markets, but there were also other formats, such as Video 2000 and U-matic. In the 1990s, digital CDs, laserdiscs and DVDs came along, followed by Blu-ray discs and streaming services in the 2000s. In this time period resolutions also changed, from high definition (HD) to ultra-high definition (UHD) and from 2k to 4k or even 8k. The array of historical technologies is a challenge for the digital historian, interested in audio-visual content. Film archives have produced digital copies from these items but there are many problems to be solved, for example, in the use of home-made movies that can still be preserved in their original format.

Visuality, textuality and metadata

As a conclusion to this chapter, I would like to return to two other aspects in the study of visuality that should be remembered and remind us of the closeness to the study of textuality. The first issue is that the dominant part of digitization projects mentioned in this book is in fact based on images. Digitized books, journals and newspapers originate from image collections, as printed material too is first scanned into an image and only thereafter is the text recognized using OCR. This means, for instance, that all newspaper repositories are basically collections of images of newspapers, although this is only seldom highlighted, as most research methods emphasize texts to which OCR has been applied. Yet, there are many other opportunities in these resources. Textuality is not the opposite of visuality, but texts are visual entities in themselves and comprise colour, font type, font size, spacing and many other layout elements. The historical change of a front page, to name only one possible theme, could be explored based on the scanned material. How did headlines change over time? How many columns were there? When did drawings and photographs first appear on the cover? How is the role of advertisements transformed if analysed in a historical setting? In addition to actual scanned images, visuality can be approached through metadata. Usually, XML files include information on the

visual contents and physical places of the texts and images on the document. This data would help the researcher to locate the drawings and photos that have been published in newspapers and magazines.

As the last aspect of this chapter, a discussion on metadata is needed, but in another sense. All databases of images contain metatextual information on each object. There is certainly variation in the quality of this data, but it will be informative in multiple ways. Let us take the example of the online databases of art museums. Tretyakov Gallery in Moscow has one of the most famous collections in the world, including over 180,000 works of art. The online collection includes the name of the artist and painting, the measurements of the artwork and the technique used for the work.[32] These are standard details, but they give the option for the researcher to explore the proportions of the collection, for example, how particular techniques change over time. Some databases consist of many more details. Let me continue with an example presented already earlier in this chapter. One of the most comprehensive databases on European art is RMN Photo, which began in 1946 as a commercial institution under the authority of the French Ministry of Culture. Its original idea was to promote the collections of the national museums of France, and it is continuing this mission; although it offers access to some foreign collections as well. Today, it comprises almost 800,000 photos of artworks housed in national and regional museums, including the Louvre, the Musée d'Orsay, the Centre Pompidou, Chantilly's Condé Museum and Palais de Beaux Arts de Lille. All these images are accessible online, although high-resolution copies can only be purchased.[33] The metadata of each item has been enriched with many details, including keywords. For example, the painting *Pauline, femme de Sénèque, rappelée à la vie* by Jean-Joseph Taillasson (1745–1809) is accompanied with an identifier (91-002122), inventory number (INV8081), period (eighteenth century, modern age), technique and material (oil on canvas), location (Paris, Musée du Louvre), credit for the photo (RMN-Grand Palais, Musée du Louvre/Christian Jean), size of the digital image (in this case 10535 × 8192 pixels) and keywords that relate to the content of the painting. This artwork portrays a fainting figure who is obviously the protagonist of the

picture, Pauline. With the keyword 'fainting', the user can detect other images of fainting. The database shows that RMN Photo covers seventeen other fainting images from the eighteenth century. In addition, the collection includes 178 artworks that portray fainting, most of them oil paintings, but there are also photographs, textiles, pencil drawings, ink works and so on. This example in itself is only one case for close reading, but if this whole database could be analysed, it would provide an exceptional view into the development of art, as well as allow other kinds of more specific research settings, such as the history of fainting.

Visual and audio-visual sources can be productively approached through metadata. This requires access to the database as a data dump. One of the most comprehensive international filmographies available is the Internet Movie Database (IMDb). It has been criticized for its incompleteness and inaccuracies, but its great benefit is its truly transnational and transcontinental coverage, not only of films made in the past but also new productions that are under preparation. IMDb was launched in 1990 as a fan-based enterprise, but today it is a subsidiary of Amazon and delivers online massive amounts of content on films, television programmes and video games. By January 2020, it included 6.5 million titles, including episodes of television series, and 10.4 million personality entries. For researchers, it is valuable that this data is largely made available through open access. IMDb offers seven different data sets that are refreshed on a daily basis.[34] These data have already been used for historically oriented studies. For example, Bo McCready has visualized the popularity of film genres between 1910 and 2018 and described the development of twelve genres, from action films to western movies.[35] This is obviously only a scratch on the surface of what could be done and achieved. The database could be employed in analysing the networks of actors, directors and other production staff; movements of people by connecting movie titles with production countries; changes in production volumes on a transregional and global level; and so on. For a film scholar, this would also offer the possibility to glimpse into the future, as the database also covers on-going productions. This sketch on the resources of IMDb is of course only one option out of many for writing

an audio-visual history. It could later be joined with other databases that include more information on national cinemas and television broadcasts. Finally, the content itself, actual films and television programmes are increasingly available in digital format.[36]

Digital technologies of visualization have been, and continue to be, an expanding area of research and development in the 2000s. Visual devices penetrate our present world, from digital cameras and mobile phones to UHD television sets and high-resolution displays. The use of these technologies has taken a central position in how we entertain ourselves, in how we express our relationship with the world around us, and how we are controlled and policed. During recent decades, audio-visual culture has undergone a transformation – from analogue movie theatres to 4k digital cinemas – to such an extent that nitrate and cellulose acetate films need special arrangements and old technology to be able to be shown. Consumers today produce visual and audio-visual content more than ever before, and they share these visualizations on social media and preserve many more on their private hard drives, as well as on cloud services. To be sure, the years to come will bring forth new insights, methods and applications that interpret history both in visual terms and based on visual cultural heritage.

4

Interdisciplinarity: Challenges for Research

Vast resources are available for the digital historian. As the previous discussions show, at the same time as more material is easily available, there are many heuristic and source-critical aspects that have to be taken into consideration. A major challenge is the use of large data sets. Therefore, digital history often appears as interdisciplinary work and takes the form of collaborative research based on cooperation and joint work. There is a need for transgressing disciplinary borders not only because of the complex problems awaiting resolution but also because, in other fields of research, such generic tools have been, or might have been, developed and productively deployed that can be applied in other contexts. Digital historians think about their work in terms of their interest in historical problems, but already problem-solving might require expertise from different disciplines and methodological traditions. It would be unfortunate not to take advantage of what has been found in other scientific communities and within different research paradigms.

Digital history does not stand alone before this challenge. The world has always been complex, to be sure, but we are increasingly conscious of the global hazards we confront, which further emphasizes the need for interdisciplinary work. In his compelling book *We Have Never Been Modern*,